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Homework 4

1) Identify two common techniques used in fault injection attacks and briefly explain how they disrupt system functionality.

Two common techniques used in fault injection attacks are clock glitching and voltage glitching. Clock glitching works by introducing abnormal clock pulses or disturbing the timing of the clock signal while the device is executing. This can cause timing violations in the hardware, which leads to skipped instructions, faulty branching or incorrect control flow. Voltage glitching involves the process of briefly lowering or spiking the supply voltage during a critical operation. Since digital circuits rely on stable voltage levels, this disturbance can cause the system to have corrupted reads, incomplete writes, bit flips or other faulty computations. In both cases, the goal ultimately is to force the device into making mistakes that can be exploited by an attacker (Bhunia & Tehranipoor, 2019).

2) What is the primary goal of a test-oriented attack in a cybersecurity context? Provide an example scenario.

The primary goal of a test-oriented attack is to exploit the built-in tests or debug the infrastructure in hardware. This allows the attacker to gain unauthorized access to sensitive information or the internal system state. These features are entirely designed for testing and validation, so they are often left unprotected and can be abused by attackers. For example, an attacker may use a scan chain to read out internal register values from a chip. If those registers

contain cryptographic keys or confidential data then the attacker can extract them without even needing to break the encryption algorithm itself. In this case, the test infrastructure itself now becomes the attack vector (Bhunia & Tehranipoor, 2019).

3) Describe how fault injection attacks can be used to bypass security measures.

Fault injection attacks can bypass security measures by intentionally disturbing the hardware during critical operations so that the system behaves incorrectly. If a clock or voltage glitch is applied during an authentication check it can skip system instructions or incorrectly evaluate the conditions allowing the attacker unauthorized access. Faults can also be introduced when the system is doing cryptographic operations. The attacker does this by producing faulty outputs in a way that can be analyzed to recover the secret keys. Instead of directly breaking the security mechanism the attacker will force the system to fail in a way that weakens or bypasses the protections that are in place (Bhunia & Tehranipoor, 2019).

4) Why might an attacker prefer using fault injection attacks over traditional hacking techniques?

Some attackers prefer using fault injection attacks because it targets the physical implementation of a system rather than relying on software vulnerabilities. Software flaws can later be patched but when the flaw is on the hardware-level it is significantly more difficult to fix once it's deployed. Additionally, fault injection attacks can bypass strong cryptographic protections without needing to mathematically break them. This makes them highly effective and desirable against strong secure systems, where traditional software-based attacks may fail. Fault injection attacks provide the attacker an alternative attack path when no obvious software vulnerabilities are present (Bhunia & Tehranipoor, 2019).

References

Bhunia, S., & Tehranipoor, M. H. (2019). *Hardware security: A hands-on learning approach*.

Morgan Kaufmann.